Branch Coverage Report

Vehicle Parking System | Team 10 | Software Engineering Mini Project

Branch coverage of the app package, measured with coverage.py --branch while running the full pytest suite

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Build tested	main branch, Team10-VPS/vehicle_parking_system (zip snapshot)
Environment	Python 3.12, Flask 3, Flask-SQLAlchemy, SQLite, pytest 9, coverage 7

1. Summary

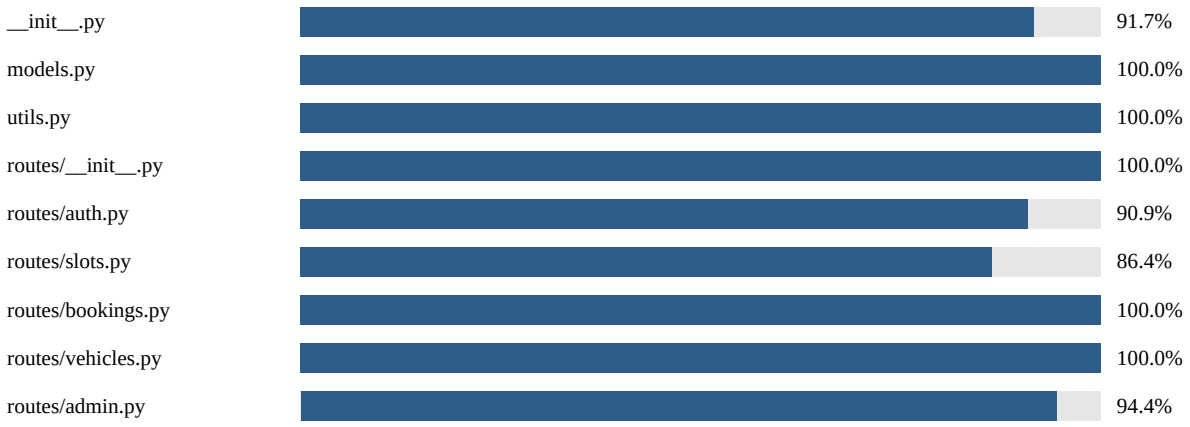
Branches	116
Covered	106
Missed	10
Partial branches	10
Branch coverage	91.4%
Statement + branch (combined) coverage	93.3%

Command used: `coverage run --branch --source=app -m pytest` then `coverage report -m`.

2. Branch coverage by file

File	Branches	Covered	Partial	Branch cov.	Missed branch arcs
app/__init__.py	12	11	1	92%	55->58
app/models.py	0	0	0	n/a	-
app/utils.py	6	6	0	100%	-
app/routes/__init__.py	0	0	0	n/a	-
app/routes/auth.py	22	20	2	91%	17->48, 81->87
app/routes/slots.py	44	38	6	86%	31->32, 34->35, 78->79, 99->100, 101->102, 117->118
app/routes/bookings.py	6	6	0	100%	-
app/routes/vehicles.py	8	8	0	100%	-
app/routes/admin.py	18	17	1	94%	62->63

Branch coverage per file (files with no branches shown as 100%)



3. Missed branches

File	Branch (from line -> to line)	Meaning
app/__init__.py	55 -> 58	config is empty: app created without a config dict (as run.py does) is not tested
app/routes/admin.py	62 -> 63	remove slot: slot id does not exist (404)
app/routes/auth.py	17 -> 48	register: plain GET request that only shows the form is never tested
app/routes/auth.py	81 -> 87	logout when nobody is logged in
app/routes/slots.py	31 -> 32	booking page: slot id does not exist
app/routes/slots.py	34 -> 35	booking page: slot already occupied
app/routes/slots.py	78 -> 79	booking: vehicle_id given but vehicle not found
app/routes/slots.py	99 -> 100	booking: new plate already owned by another user
app/routes/slots.py	101 -> 102	booking: existing plate with a different vehicle type
app/routes/slots.py	117 -> 118	booking: slot disappears before commit

4. Conclusion

Branch coverage is 91.4% (106 of 116 branches). Most of the missed branches are in slots.py and admin.py error handling, matching the lines listed in the Code Coverage Report. Adding tests for a plain GET on /register, logout without a session, and the missing or occupied slot cases on the booking page would close most of the gap.